GENAI Pair Programming Python — Capstone Project

Task 2: AI-Powered Data Cleaning Assistant

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Architecture: Modular Pipeline Engine	**Dependencies:** 100% Standard Library

1. Executive Summary & Capabilities

The AI-Powered Data Cleaning Assistant is an enterprise-grade tabular data preprocessing engine designed to clean messy CSV datasets automatically without external dependencies.

Pipeline Stage	Methodology & Logic
1. Type Inference & Sanitization	Parses currencies ($1,200.50), normalizes ISO dates (YYYY-MM-DD), converts null tokens ('N/A', 'missing') to None.
2. Insertion-Order Deduplication	Preserves original row order while filtering duplicate records.
3. Missing Value Imputation	Applies column-wise statistical mean/median for numerical data and mode for categorical data.
4. Outlier Winsorization Capping	Computes 1.5 * IQR bounds (Q1, Q3) to cap extreme values.
5. Structured Telemetry & Audit	Logs delta row metrics per step and outputs Markdown audit summary.

2. Core Engine Implementation (Extract)

```
class DataCleaningAssistant:
    def clean_dataset(self, headers: list[str], rows: list[list[Any]]) -> tuple[list[list[Any]], Clean
ingAuditReport]:
        # 1. Type Sanitization & Null Identification
        typed_rows = [[infer_and_cast_value(val) for val in row] for row in rows]
        # 2. Insertion-Order Deduplication
        deduped_rows = deduplicate_matrix(typed_rows)
        # 3. Statistical Missing Value Imputation (Mean/Median/Mode)
        imputed_rows = impute_missing_matrix(headers, deduped_rows)
        # 4. Outlier Capping via IQR Winsorization
        final_rows = cap_outliers_matrix(headers, imputed_rows)
        return final_rows, report
```

3. Verification & Test Execution Results

All 40 unit and integration tests across the test suite (test_algorithms, test_data_cleaner, test_data_transformer, test_output_handler, test_e2e) passed with 100% OK rating.